

SMART SHOCK DETECTION & AUTOMATIC POWER CUTOFF SYSTEM

Preventing Electrocution During Floods and Waterlogging

IDEA LAB WORKSHOP PROJECT REPORT

Submitted By

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Academic Year: 2025–2026

Certificate

This is to certify that the project titled “**Smart Shock Detection & Automatic Power Cutoff System**” submitted by **Mahbub Ahmed Barbhuiya, Ashmrit Das, Adnan Sajjad Mazumder** and Mitu Patowary for **Idea Lab Workshop** in the Department of **Electrical Engineering** at **Assam Engineering College** is a record of bonafide work carried out during the academic year 2025–2026.

The project demonstrates the implementation of an intelligent electrical safety system for detecting leakage current and automatically disconnecting power supply during flood and waterlogging conditions. The students have successfully completed the project work as per the academic requirements.

We hereby approve the project report for submission and evaluation.

Project Guide Signature

HOD Signature

Acknowledgement

We would like to express our sincere gratitude to everyone who supported and encouraged us throughout the completion of this project titled “Smart Shock Detection & Automatic Power Cutoff System”.

We are deeply thankful to the faculty members and the Department of Electrical Engineering, Assam Engineering College, for providing us with the guidance, facilities, and encouragement required for completing this project successfully. Their support helped us gain practical knowledge about embedded systems, automation, and electrical safety technologies.

We also extend our gratitude to our friends and classmates for their cooperation, suggestions, and continuous motivation during the development of the project, and sincerely thank everyone who contributed directly or indirectly towards the successful completion of this work.

Abstract

Floods and heavy rainfall create serious electrical hazards in waterlogged areas due to exposed wires, damaged electric poles, underground cables, and leakage currents. During such situations, electricity may continue flowing through flooded regions, creating a high risk of accidental electrocution for pedestrians, residents, workers, and domestic animals. Every year, many lives are affected due to the absence of automatic electrical safety systems in flood-prone areas.

The proposed **Smart Shock Detection & Automatic Power Cutoff System** is designed to improve public safety by automatically detecting dangerous leakage current and disconnecting the power supply before accidents occur. The system uses current sensors and water level sensors to continuously monitor flooded areas. The sensor data is processed by a microcontroller such as Arduino UNO or ESP32, which controls a relay module for automatic power cutoff during dangerous conditions.

The system also includes warning features such as buzzers and LED indicators to alert nearby people about electrical danger. The project demonstrates the practical implementation of embedded systems, automation, sensor interfacing, and electrical protection technologies for solving real-life safety problems.

The proposed solution is low-cost, reliable, easy to implement, and suitable for flood-prone regions, public infrastructure, and smart city safety applications. Overall, the project aims to reduce electrocution accidents and enhance public electrical safety during floods and heavy rainfall conditions.

The project demonstrates automation, embedded systems and electrical protection technologies for improving public safety.

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1 Introduction

Floods and heavy rainfall are very common in India during the monsoon season. Water often comes into contact with damaged electrical wires and underground cables.

This creates leakage current in flooded areas and increases the risk of accidental electrocution.

The Smart Shock Detection & Automatic Power Cutoff System automatically detects dangerous leakage current and disconnects the power supply immediately.

The system is designed to be:

- Low cost
- Reliable
- Easy to implement
- Fast responding
- Suitable for flood-prone areas

2 Identifying Problem

Several major problems were identified during floods:

- Exposed live wires
- Leakage current in standing water
- Broken electrical poles
- No automatic protection system
- High electrocution risk

An intelligent and automatic electrical safety system is therefore required.

3 Objective

The main objective of the project is to develop a smart safety system that reduces electrocution risks during floods and waterlogging situations.

Specific objectives include:

- Detect electrical leakage current
- Monitor dangerous conditions continuously
- Automatically disconnect power supply
- Alert nearby people
- Improve public safety

4 Brainstorming

Different technologies were explored during planning:

- Current sensing systems
- Water level sensors
- IoT safety systems
- Relay protection systems
- GSM alert systems

The final system architecture selected was:

Sensors → Arduino Processing → Alert System → Relay Cutoff

5 Concept Design

The complete system consists of:

- Leakage Detection Unit
- Water Detection Unit
- Processing Unit
- Alert Unit
- Relay Protection Unit

5.1 Working Principle

The operation sequence is:

1. Sensors monitor leakage current
2. Arduino receives sensor data
3. System checks danger level
4. Relay disconnects power supply
5. Buzzer and LEDs generate warning

5.2 Components Used

- Arduino UNO / ESP32
- ACS712 Current Sensor
- Water Level Sensor
- Relay Module
- Buzzer
- LEDs
- LCD Display

5.3 Role of Relay Module

The relay module acts as an electrically controlled switch that disconnects power supply during dangerous conditions.

6 Prototype

A prototype model of the system was developed using simulation tools and electronic components.

6.1 System Workflow

The workflow includes:

- Sensor Detection
- Arduino Processing
- Alert Generation
- Safety Action

6.2 Hardware Connections

Current Sensor Connections

- VCC connected to 5V
- GND connected to GND
- Output connected to Arduino pin

Relay Module Connections

- VCC connected to 5V
- GND connected to GND
- IN connected to digital pin

7 Testing

The system was tested under different flood and leakage conditions.

7.1 Leakage Detection Test

- Current sensor detected abnormal current
- Arduino processed danger condition
- Red LED turned ON
- Buzzer activated

7.2 Automatic Cutoff Test

- Relay disconnected power supply
- Warning message displayed
- Protection mode activated

8 Implementation

The proposed system can be implemented in:

- Flood-prone roads
- Residential colonies
- Railway stations
- Construction sites
- Industrial zones

9 Advantages

- Reduces electrocution accidents
- Automatic operation
- Fast emergency response
- Low-cost implementation
- Reliable protection

10 Limitations

- Sensor accuracy may be affected
- Large areas require multiple sensors
- Battery backup is necessary
- Outdoor systems require weatherproofing

11 Future Scope

Future improvements include:

- IoT-based monitoring
- AI-based danger prediction
- GPS tracking system
- Cloud data storage
- Mobile application integration
- Automatic SMS alert system

12 Expected Outcome

Expected results include:

- Reduction in electrocution accidents
- Faster emergency response
- Automatic power isolation
- Better safety monitoring

13 Conclusion

The **Smart Shock Detection & Automatic Power Cutoff System** is an innovative and practical solution developed to reduce the risk of electrocution during floods and waterlogging situations. The project successfully demonstrates how automation and embedded systems can be used to improve public electrical safety in flood-prone areas.

The system continuously monitors leakage current and dangerous electrical conditions using sensors and automatically disconnects the power supply through a relay module whenever a threat is detected. The inclusion of warning indicators such as buzzers and LEDs further helps in alerting nearby people and preventing accidents.

This project combines important technologies such as sensor interfacing, microcontroller programming, relay protection, and automation to create an efficient and reliable

safety system. The proposed solution is low-cost, easy to implement, and suitable for practical applications in public infrastructure, residential areas, industrial zones, and smart city systems

Overall, the project highlights the importance of intelligent electrical protection systems and demonstrates a meaningful technological approach towards improving public safety during natural disasters and emergency situations.